

Import Data using Transform Maps (Spreadsheet)

Description

This micro project focuses on importing structured data from an external spreadsheet into the ServiceNow platform using Import Sets and Transform Maps. The project simulates a real-world scenario where bulk employee data is received in Excel format and must be migrated into ServiceNow efficiently and accurately. The spreadsheet data is first loaded into an Import Set table, which serves as a staging area, and a Transform Map is then configured to map source fields to the appropriate target table fields. Upon executing the transformation, records are created in the target table and validated to ensure data integrity.

Step by Step Process

Phase 1: Prepare the Sheets

1. Open New Google Spreadsheet.
2. Create a new Google Spreadsheet with sample data.
3. Download the spreadsheet as Sample Spreadsheet.xlsx format.

Phase 2: Create New Custom Table

ServiceNow • Example Screenshot

Application Navigator

All

Tables

Load Data

Transform Maps

Reports

Dashboards

Create New Table

Label

Employee Test

Name

u_employee_test

Table type

Custom

Save

Example only — UI may vary by ServiceNow version.

Application Navigator

All

Tables

Load Data

Transform Maps

Reports

Dashboards

Form Layout — Add Field

Column Label

Employee ID

Type

String

Column Label

Employee Name

Type

String

Add

Save

Example only — UI may vary by ServiceNow version.

Navigation: Tables > Create New

- Label: Employee Test
- Name: u_employee_test
- After creating and saving the Employee Test table, scroll down to the bottom of the table form.
- Click Show Form to open the table form.
- Open Form Context Menu / Additional Actions on the form header.
- Click Configure and select Form Layout.
- In Form Layout, click New to create a new field.
- Column Label: Employee ID
- Type: String
- Click Add to create the field.
- Repeat for Employee Name, Email, Department, and Location. Each field is Type: String.
- Click Save and verify that the new fields are displayed on the Employee Test form.
- Click Submit or Save.

Phase 3: Import Set Table

ServiceNow • Example Screenshot

Application Navigator

All

Tables

Load Data

Transform Maps

Reports

Dashboards

Load Data

Import set table

Employee Import

File

Sample Spreadsheet.xlsx

Sheet number

1

Header row

1

Submit

Example only — UI may vary by ServiceNow version.

Navigation: Application Navigator > Load Data

- Click Load Data and create an import table.
- Label: Employee Import
- Name: u_employee_import (auto populated)
- Click Submit.
- Verify the success message.
- Click Create Transform Map.

Phase 4: Create Transform Map

ServiceNow • Example Screenshot

Application Navigator

All

Tables

Load Data

Transform Maps

Reports

Dashboards

Create Transform Map

Name

Sample Spreadsheet Import

Source Table

Employee Import

Target Table

Employee Test

Auto Map

Mapping Assist

Save

Example only — UI may vary by ServiceNow version.

Application Navigator

All
Tables
Load Data
Transform Maps
Reports
Dashboards

Run Transform

Import Set

Employee Import

Transform Map

Sample Spreadsheet Import

State

Ready

Transform

Example only — UI may vary by ServiceNow version.

- Click Create Transform Map.
- Name: Sample Spreadsheet Import
- Source Table: Employee Import (auto-populated)
- Target Table: Employee Test
- Click Auto Map Matching Fields.
- Click Mapping Assist and map the source fields to the target fields.
- Click Save.
- Verify that both Source and Target table fields are mapped.
- Click Transform and then click the Transform button.
- Verify the success message after the transformation.

Phase 5: Transform Data & Validate

- Open the Employee Test table from the Application Navigator.
- After the import succeeds, the records are created in the Employee Test table.
- Rearrange the fields using Personalised List Columns.
- Click OK.
- Final result: all records are successfully imported into ServiceNow using Import Set and Transform Map.

Phase 6: Enable Coalesce to Avoid Duplicate Records

ServiceNow • Example Screenshot

Application Navigator

- All
- Tables
- Load Data
- Transform Maps
- Reports
- Dashboards

Field Map — Coalesce

Target field

Coalesce

Source field

Example only — UI may vary by ServiceNow version.

In real projects, the same employee data may be imported multiple times. Coalesce is used to prevent duplicate employee records. When duplicate values are found, the existing records are updated with new values instead of inserting duplicate records.

4. Open Transform Map under System LDAP On All menus.
5. Open the created Transform Map: Sample Spreadsheet Import.
6. Under Related List, open the Field Maps section and make any one field Coalesce = True.
7. Save the form.
8. Add some data again in Excel format.

Phase 7: Inserting New Data in Excel Format

- Navigate to All menus and search for Load Data under System Import Set.
- Open Load Data.
- Import Set Table: select Existing Table (Employee Import).
- Source of Import: Select File and choose the Excel file.
- Sheet number: 1; Header row: 1.
- Click Submit.
- The newly created Excel data contains previously added data and new/changed values.
- Example changes in the reference: Raviteja's email changes from test10@gmail.com to test18@gmail.com, and Employee ID SB-0004 name changes from Ajay kumar to Ajay.
- Two other new Employee IDs are also added.
- After submission, verify State = Completed and Success.
- Click Run Transform under the next steps.
- Confirm the previously selected Import Set and Transform Map, then click Transform.
- Verify State = Completed and Success.
- Open Transform History to review Inserted, Updated, Skipped, or Ignored values.

- In the reference example, 4 rows are uploaded: 2 are inserted as new IDs and 2 are updated because their names/emails changed.
- Open the Employee Test table and verify the updated and inserted rows.
- Repeat Phase 7 with the same Excel sheet to verify duplicate handling.
- Open Transform History again. In the reference example, 4 rows are uploaded and 0 are inserted, 0 are updated, and 4 are ignored.

Phase 8: Create Reports on Employee Test Table

ServiceNow • Example Screenshot

Application Navigator

- All
- Tables
- Load Data
- Transform Maps
- Reports
- Dashboards

Create Report — Employees by Department

Report name:

Table:

Type:

Group by:

Aggregation:

Example only — UI may vary by ServiceNow version.

Navigation: All menus > Reports > Usage and Governance > Reports

Create three reports:

Report 1: Employees by Department

- Report name: Employees by Department
- Source Type: Table
- Table: Employee Test
- Type: Pie chart
- Group by: Department
- Aggregation: Count
- Leave chart color as default.
- Click Run, then Save.

Report 2: Employees by Location

- Report name: Employees by Location
- Table: Employee Test
- Type: Bar Chart
- Aggregation: Count
- Group By: Location
- Click Run, then Save.

Report 3: Employee List Report

- Report name: Employee List Report
- Table: Employee Test
- Type: List
- Columns: Employee ID, Employee Name, Email, Department, Location
- Click Next; no Style configuration is required.
- Click Run, then Save.

Phase 9: Add Reports to Dashboard

ServiceNow • Example Screenshot

Application Navigator

- All
- Tables
- Load Data
- Transform Maps
- Reports
- Dashboards

Employee Analytics Dashboards

Dashboard name

Widgets

Example only — UI may vary by ServiceNow version.

Create Dashboard

- Open All menus and search for Dashboards.
- Under Self Service, click Dashboards.
- Click Create Dashboards or use the backend name PA_DASHBOARDS.FORM.
- Click New to create a dashboard.
- The backend form requires ACL access to pa_dashboards with Admin Overrides = True.
- Search ACLs for pa_dashboards, open the Create Operation row, enable Admin Overrides = True, and update the form.
- Open PA_DASHBOARDS.FORM.
- Dashboard name: Employee Analytics Dashboards.
- Save the form.

Add Reports to Dashboard

- Open All menus and search for Reports under Usage and Governance.
- Open Employees by Department.
- Select Edit Report, click Share, and choose Add to Dashboards.
- Choose Employee Analytics Dashboards and click Add.
- Repeat the same process for Employees by Location and Employee List Report.
- Alternatively, use View Report > Share > Add to Dashboards.

- The three reports are added successfully to the dashboard.
- Widgets can also be added using the + icon, and the dashboard background can be configured if required.
- Users, Groups, and Roles can be restricted through the Share option.

Conclusion

9. This project successfully implemented an automated employee data management solution using ServiceNow Import Sets and Transform Maps, ensuring accuracy, efficiency, and data integrity throughout the employee onboarding and update process.
10. Coalesce was enabled in the Transform Map to identify existing employee records during imports, prevent duplicate records, and update existing data instead of inserting redundant entries.
11. Reports and Dashboards were added to provide visibility into employee data and import activities, including total employee records, newly added employees, updated employee records, department-wise employee distribution, and import success/error counts.
12. The reports were consolidated into a dashboard so administrators and HR teams can monitor employee data trends, import performance, and data quality from a centralized view.

EXAMPLE SCREENSHOTS

The screenshots below are illustrative examples added to make the procedure easier to understand. They are not captured from the user's ServiceNow instance, so labels or screen placement may vary by version.